Genuine Price Anomaly Detection and Q-Learning Dispatch for Battery Storage

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Abstract—Battery energy storage system (BESS) arbitrage profits depend disproportionately on rare, high-magnitude price spikes, yet existing anomaly detection literature predominantly benchmarks on synthetic perturbations rather than authentic wholesale market settlements. This paper presents an end-to-end framework combining unsupervised deep recurrent autoencoders with reinforcement learning dispatch, evaluated across three years (25,944 hours, 2023–2025) of raw California Independent System Operator (CAISO) day-ahead prices. A Gated Recurrent Unit (GRU) autoencoder achieves 0.882 AUROC on single-hub detection (0.894 under seasonal transfer) while training 25% faster than its LSTM counterpart. Conditioning tabular Q-learning on the resulting anomaly flag quadruples six-month energy arbitrage gains to $\$11,957$ while cutting cross-seed policy variance by 90%. Under adversarial price feed poisoning, our defense classifier achieves 89.1% precision and 97.2% AUROC, mitigating attack-induced degradation by 67.2% and comfortably satisfying the break-even precision requirement of $p^* = 33.4\%$, whereas conventional static thresholds fail economically ($p^* > 100\%$). These findings demonstrate that wholesale price spikes should be exploited as profitable dispatch signals rather than discarded as corrupted data, establishing an actionable alternative to synthetic-anomaly benchmarking.

Index Terms—Adversarial robustness, anomaly detection, battery energy storage systems, California Independent System Operator, energy arbitrage, graph neural networks, peer-to-peer networks, reinforcement learning.

I. Introduction & Literature Review

Electricity prices in California have grown considerably more unpredictable in recent years. Solar generation during midday hours frequently depresses locational marginal prices (LMPs) toward zero or negative territory, only for prices to rebound sharply after sunset as gas-fired peaker plants ramp up generation to cover the evening peak. Battery energy storage systems (BESS) exploit this daily cycle in wholesale power markets and decentralized peer-to-peer (P2P) trading frameworks, charging during low-price hours and discharging as prices climb [1], [2].

Under routine operating conditions, these diurnal price trajectories follow predictable patterns driven by work schedules, weather conditions, and network loading. However, annual storage profitability is predominantly driven by a handful of extreme price events. During regional heatwaves, transmission contingencies, and sudden generator outages, hourly prices can soar above $\$1{,}000$/MWh. A 10 MW / 10 MWh battery that discharges fully during a single such spike can capture revenues comparable to an entire week of routine cycling. Detecting and anticipating these rare, high-value settlement intervals in real time is therefore of primary importance for storage operators.

Most state-of-the-art (SOTA) approaches to anomaly detection in pricing have addressed the problem primarily from a cybersecurity perspective [3]–[5]. A common experimental convention trains detectors on clean historical traces, injects synthetic price spikes, and reports detection fidelity on the artificial perturbations [6]. While informative for evaluating cyber-physical attack resilience, this formulation divorces anomaly detection from real-world market settlement mechanics. Crucially, misclassifying a legitimate $\$1{,}000$/MWh market spike as an anomaly and withholding battery discharge causes severe revenue loss.

To reflect authentic market conditions, our benchmark evaluates seven detector architectures using solely actual settlement prices retrieved directly from the California Independent System Operator (CAISO) Open Access Same-Time Information System (OASIS) [7] without synthetic modifications. We also examine the dual operational challenge: how adversarial entities might inject deceptive price spikes into telemetry feeds, and how storage dispatchers can economically defend against such manipulation.

Evaluating anomaly detectors on historical market data introduces two fundamental challenges. First, historical wholesale settlements lack objective ground-truth anomaly labels, requiring validation across both statistical robustness metrics and downstream economic profitability. Second, classification errors exhibit extreme financial asymmetry. Whereas a false positive in conventional time-series filtering is merely an annoyance, suppressing discharge during a genuine $\$1{,}000$/MWh spike imposes irreversible financial penalties.

This work sits at the intersection of electricity price forecasting [8], unsupervised time-series anomaly detection [4], [9], generative modeling [10], [11], P2P energy trading [1], [2], and battery dispatch optimization [12]–[15].

Electricity Prices and Anomaly Detection: Rolling statistics, moving averages, and local z-scores are traditional approaches for detecting abnormal price levels in SOTA models. While computationally straightforward, static thresholding struggles with strong seasonality: a threshold calibrated to winter baselines triggers spurious false alarms during summer demand peaks.

The deep models learn the usual 24-hour price pattern directly from historical data; there are no predefined patterns in our models. The recurrent autoencoders (Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU)) transfer the sequence states forward through time,

compress each daily path to a small bottleneck, and detect moments when the reconstruction error becomes large [9]. On the other hand, the variational autoencoders (VAE) encode price windows into a continuous latent space [10], and the denoising diffusion probabilistic models (DDPM) add Gaussian noise to the input and learn to reverse this process [11]. Large price spikes are unusual and harder to denoise, leading to elevated residuals.

Battery Dispatch and Trading: In storage scheduling, commonly used methods include model predictive control (MPC), linear programming, and dynamic programming [13], while reinforcement learning (RL) is increasingly seen as an alternative to these methods [1], [12], [15]. A recurring shortcoming in existing dispatch literature is the assumption of perfect price foresight; when the controller possesses advance knowledge of clearing prices, anomaly detection becomes redundant. Our formulation eliminates lookahead bias by enforcing strict causality across all dispatch policies.

The reasons behind these two problems, lack of labeled data and imbalanced misclassification costs, inspire the design choices made in this paper. The paper is valuable in the following three aspects:

1) First, we perform a benchmarking of seven different detector architectures—Isolation Forest (iForest), a Variational Autoencoder (VAE), a denoising diffusion model, single-node LSTM and GRU autoencoders, a multivariate LSTM, and a graph attention network—on 25,944 hours of actual CAISO data, across five random seeds and two disjoint test windows, in order to find out which one performs best in detecting price spikes versus everyday diurnal fluctuations.
2) On top of the detection signal obtained from these tests, we then apply rules-based, directional, and reinforcement learning-based dispatch policies to understand how different types of dispatchers exploit the signal generated by the detectors to make decisions that yield the best results. We prove that tabular Q-learning achieves maximum exploitation of the anomaly-detection signal, leading to nearly a fourfold increase in arbitrage profits.
3) Furthermore, the introduced detector can be targeted by price feed poisoning attacks. To assess the economic feasibility of defense from an asymmetric-risk perspective, we introduce the economic break-even point precision p^* and prove that threshold-based defenses fail economically ($p^* = 1.017 > 100\%$). On the contrary, our approach reaches the break-even point at 89.1% precision ($p^* = 0.334$), mitigating up to 67.2% of the attack losses.

II. Mathematical Modeling

To formalize the physical battery constraints, detection metrics, RL dynamics, and adversarial risk, nomenclature and parameter settings are presented in Table I.

TABLE I
Nomenclature and Parameter Settings

Symbol	Description
s_t	Anomaly score for the 24h window ending at hour t
H	Window length ($H = 24$ hours)
$x_{t,h}, \hat{x}_{t,h}$	Normalized observed and reconstructed price at hour h
$Q(s,a)$	Action-value estimate for state s and action a
α, γ	Q-learning rate ($\alpha = 0.15$) and discount factor ($\gamma = 0.95$)
r_t	Net arbitrage cash return at hour t ($)
SOC_t	Battery energy level at hour t ($0 \le SOC_t \le 10$ MWh)
$C, P_{\max}$	Rated energy capacity (10 MWh) and power limit (10 MW)
$e, \Delta t$	Round-trip cycle efficiency ($e = 0.80$) and interval ($\Delta t = 1$ h)
$c_{\text{fee}}, c_{\text{deg}}$	Market fee ($0.50/MWh) and cell degradation ($2.50/MWh)

Discrete-time state-of-charge (SOC) dynamics govern the physical operation of the BESS:

$$SOC_{t+1} = SOC_t + \left(\eta_{\text{ch}} P_t^{\text{ch}} - \frac{1}{\eta_{\text{dis}}} P_t^{\text{dis}}\right) \Delta t, \quad (1)$$

subject to operational power rating and storage capacity limits:

$$0 \le P_t^{\text{ch}} \le P_{\max}, \quad 0 \le P_t^{\text{dis}} \le P_{\max}, \quad (2)$$

$$0 \le SOC_t \le C, \quad (3)$$

where P_t^{ch} and P_t^{dis} denote power at hour t, respectively, with $P_{\max} = 10$ MW, $C = 10$ MWh, and $\Delta t = 1$ h. Symmetric one-way conversion efficiencies satisfy $\eta_{\text{ch}} = \eta_{\text{dis}} = \sqrt{e} = \sqrt{0.80} \approx 0.894$, guaranteeing a round-trip cycle efficiency of $e = \eta_{\text{ch}}\eta_{\text{dis}} = 0.80$.

The deep sequential model considers normalized 24-hour sliding price vectors and evaluates their anomaly score by means of mean squared reconstruction error:

$$s_t = \frac{1}{H} \sum_{h=1}^{H} \left(x_{t,h} - \hat{x}_{t,h}\right)^2. \quad (4)$$

Usually, the autoencoder successfully reconstructs the price graph for the particular day. However, when prices increase suddenly, the compressed representation is unable to encode the change, and s_t increases sharply.

In the GNN-AE model, we use the graph representation $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ for the three regional hubs, where $\mathcal{V} = \{\text{NP15}, \text{ZP26}, \text{SP15}\}$, and the edge set $\mathcal{E}$ contains the main transmission connections for Path 15 (between NP15 and ZP26) and Path 26 (between ZP26 and SP15). In the graph attention network, two graph attention layers are used (hidden units of 8), and the directional spatial attention weight α_{ij} is computed as:

$$\alpha_{ij} = \frac{\exp\left(\text{LeakyReLU}\left(\mathbf{a}^T[\mathbf{W}h_i \parallel \mathbf{W}h_j]\right)\right)}{\sum_{k\in\mathcal{N}_i} \exp\left(\text{LeakyReLU}\left(\mathbf{a}^T[\mathbf{W}h_i \parallel \mathbf{W}h_k]\right)\right)}, \quad (5)$$

where $\mathbf{W} \in \mathbb{R}^{8\times d}$ is a shared projection matrix, $\mathbf{a}$ is the attention vector, $\parallel$ is concatenation, and $\mathcal{N}_i$ denotes

neighboring regional hubs. Spatial node representations are followed by LSTM-based temporal sequence encoding.

Tabular Q-learning [15] learns to dispatch in a discrete state space defined by $s = (p_z, a_{\text{flag}}, s_{\text{soc}})$. The z-normalized price is partitioned into $N_p = 10$ discrete bins over $[-3, +3]$; the anomaly flag is mapped to $N_a = 5$ categories (or $N_a = 1$ for the unconditional baseline); and the battery state of charge (SOC_t/C) is divided into $N_s = 5$ bins, yielding $|S| = 250$ discrete state representations ($|S| = 50$ without the anomaly flag). The action space contains three discrete power actions $a \in \{-1.0, 0.0, 1.0\}$ corresponding to full discharge ($P_t^{\text{dis}} = P_{\max}$), idle ($P_t = 0$), and full charge ($P_t^{\text{ch}} = P_{\max}$), subject to the physical constraints in (1)–(3). The net cash return reward r_t received by the agent accounts for market revenue, degradation, and fees [16]:

$$\begin{aligned} r_t = &\left(p_t P_t^{\text{dis}} - p_t P_t^{\text{ch}}\right) \Delta t \\ &- \left[c_{\text{fee}}\left(P_t^{\text{dis}} + P_t^{\text{ch}}\right) + c_{\text{deg}} P_t^{\text{dis}}\right] \Delta t. \end{aligned} \tag{6}$$

The action values are iteratively learned using Eq. (7):

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left[r_t + \gamma \max_{a'} Q(s',a') - Q(s,a)\right], \tag{7}$$

with learning rate $\alpha = 0.15$ and discount factor $\gamma = 0.95$.

III. Methodology

The experimental evaluation utilizes three years of hourly day-ahead LMPs from CAISO spanning January 1, 2023, to December 31, 2025. Settlement traces were retrieved for three major trading hubs (SP15, NP15, and ZP26) alongside four ancillary service products (Regulation Up, Regulation Down, Spinning Reserves, and Non-Spinning Reserves), totaling 25,944 hourly observations per stream. The ancillary service price streams provide auxiliary contextual signals for multi-channel detection, reflecting system-wide reserve scarcity. Economic dispatch and arbitrage evaluations focus strictly on day-ahead energy market settlements.

Substantial price volatility occurred across the historical window, with SP15 reaching a peak of $\$1,143.78$/MWh on August 17, 2023, during regional heatwaves and falling to $-\$41.33$/MWh on May 19, 2024, during midday solar overgeneration.

To prevent data leakage, the evaluation timeline is partitioned into disjoint chronological intervals: a training split from January 12, 2023, to December 31, 2024 (17,272 hours, approximately 23.7 months), a validation split from January 1 to June 30, 2025 (4,272 hours, 5.9 months, used exclusively for threshold calibration), and an out-of-sample test split from July 1 to December 31, 2025 (4,400 hours, 6 months, evaluated once). We additionally evaluate an out-of-sample seasonal test split spanning January 1 to June 30, 2025. In California wholesale markets, Day-Ahead Market (DAM) clearing occurs for all 24 hours of the subsequent operating day at 13:00 $D-1$. However, commercial BESS traders must submit energy schedules prior to market close at 10:00 AM without advance knowledge of settled clearing prices, subsequently managing real-time exposure across 15-minute and 5-minute intervals. All models operate strictly on historical trailing 24-hour windows to enforce causality and eliminate lookahead bias.

Because objective ground-truth anomaly labels do not exist in historical market records, a statistical benchmark label is defined strictly for ex-post evaluation. An observation is designated as anomalous if its clearing price exceeds 4 median absolute deviations (MAD) from the 30-day trailing rolling median or falls within the extreme 0.5% tails of historical settlement prices, identifying 517 anomalous hours across 25,944 observations (approximately 2.0% of the timeline). No detector has access to these labels during training. Sensitivity analysis confirms that relative performance rankings across detector families remain invariant to the choice of deviation threshold (evaluated across 3.0 to 5.0 MAD).

The detection pipeline (Fig. 1) uses a two-level filtering approach: iForest is used to isolate and remove major outliers, followed by analysis of the remaining possible hours using a deep-sequence model.

The Stage 1 iForest [17] considers nine historical-only causal features: (i) trailing 6-hour linear price slope $\Delta p/\Delta t$, (ii) non-reversion ratio $|p_t - p_{t-1}|/(|p_{t-1} - p_{t-2}| + \epsilon)$, (iii) 1-hour price difference $(p_t - p_{t-1})$, (iv) 3-hour price difference $(p_t - p_{t-3})$, (v) 24-hour trailing robust z-score $(p_t - \text{median}_{24})/\text{MAD}_{24}$, (vi) 168-hour trailing z-score, (vii) 24-hour rolling volatility σ_{24}, and (viii–ix) cyclic diurnal harmonics $(\sin(2\pi h/24), \cos(2\pi h/24))$ for hour-of-day $h \in \{0, \ldots, 23\}$.

A total of three models have been implemented for Stage 2 single-node detection. VAE has been trained for 40 epochs with the Adam optimizer (lr $= 10^{-3}$, batch size=64). The VAE architecture consists of fully connected networks with 48 neurons, and the bottleneck is modeled as an 8-dimensional Gaussian distribution [10]. LSTM-AE and GRU-AE map the price sequence through 32 recurrent neurons to an 8-dimensional bottleneck.

The neural network is trained using the Adam optimization algorithm, with a learning rate of 10^{-3}, a batch size of 32, and 40 training epochs. The training uses the mean squared error loss function [9]. In the GRU case, the input and forget gates are used, which makes training 25% faster but does not affect temporal dependencies. The DDPM learns how to reverse the procedure of adding noise to the data on discrete schedules [11]. The inference step involves corrupting the window samples at four noise levels ($t \in \{0.10, 0.25, 0.40, 0.60\}$) over 16 iterations and averaging the residuals.

To explore the influence of the regional transmission environment on predictability, we also feed SP15, NP15, and ZP26 as inputs to two multi-hub models, which include a 3-channel multivariate LSTM (MV-LSTM) control and the graph neural network autoencoder (GNN-AE) [18].

All experiments were conducted using five random seeds (0 to 4), ensuring that torch.set_num_threads(1) was always set and that cuDNN flags were fixed to ensure

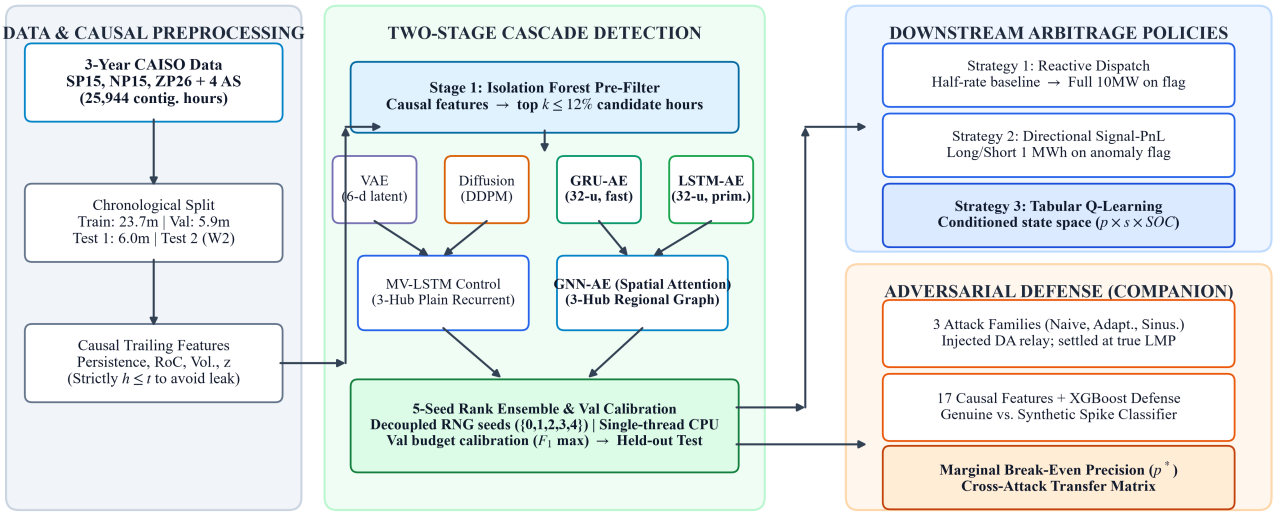


Fig. 1. System architecture. The raw CAISO price data undergoes causal feature extraction before Stage 1, which consists of an iForest pre-filter, followed by Stage 2, which consists of deep autoencoders and multi-node models. The five-seed ensemble then guides the battery dispatching policy and the defense filter.

determinism. To evaluate the performance of the proposed framework, all algorithmic evaluations were executed on a commodity laptop host consisting of a Dell Inspiron 15 workstation. The hardware execution layer features an Intel Core i7-1355U processor (10 cores, 1.70 GHz base clock, up to 5.0 GHz boost configuration), 16 GB of system RAM, and integrated Intel Iris Xe Graphics (128 MB). High-throughput dataset input/output operations were sustained using a 512 GB NVMe SSD operating under a Microsoft Windows 11 Home Single Language environment.

The core computational pipelines were implemented within a unified Python ecosystem utilizing numpy (v2.0+) for matrix and tensor computations, pandas (v2.2+) for settlement series indexing, and scipy (v1.14+) for linear programming via the HiGHS solver. Machine learning implementations and defensive modeling relied on scikit-learn (v1.5+) for Isolation Forest and validation metrics, alongside torch (v2.4+) for deep autoencoders and Graph Neural Networks (GNNs). Finally, gradient boosted defenses were deployed via xgboost (v2.0+), with all performance and data visualizations generated using matplotlib (v3.8+).

IV. Results and Experimental Evaluation

Thresholds for the budget percentile range from 0.5% to 10% were optimized based on the F1 score on the validation dataset and fixed prior to evaluating the model on the test dataset.

The results for the quality of reconstruction and detection of anomalies by all methods are provided in Table II, measured in terms of root mean squared error (RMSE) and mean absolute error (MAE). The ROC curves are shown in Fig. 2, highlighting the detection trade-off across candidate model families.

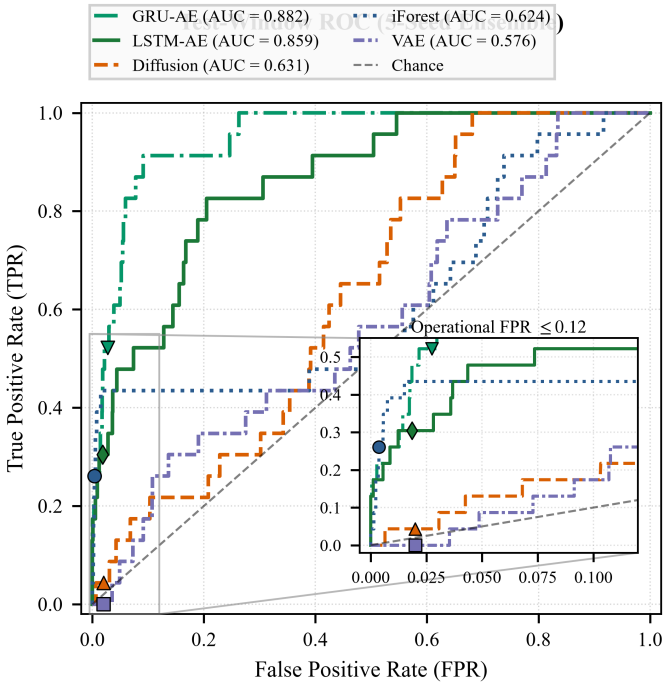


Fig. 2. Test-window ROC curves for the 5-seed ensemble models on CAISO data, GRU-AE (downward triangle), LSTM-AE (diamond), Diffusion (upward triangle), iForest (circle), VAE (square), with the inset showing the operational low-false-positive region (false positive rate FPR ≤ 0.12).

Perhaps the most obvious trend from Table II is that low reconstruction error does not necessarily indicate high-quality anomaly detection. While the VAE yielded the highest-quality reconstructions out of all single-node models (RMSE \$7.85/MWh), it generated the lowest detection scores (AUROC = 0.534; $F_1 = 0.027$). The VAE relies on the latent prior to generate Gaussian noise, so the

TABLE II
Performance Master Summary: Accuracy of Model Reconstruction and Anomaly Identification

	Reconstruction Error (↓)		Anomaly Detection Accuracy (↑)				Operating Point Metrics (↑)		
Model Architecture	RMSE ($)	MAE ($)	Mean AUROC	Ens. AUROC	Avg. Prec.	W2 AUROC	F_1 Score	Precision	Recall
Single-Node Models (Evaluated on CAISO SP15 Hub)									
iForest	n/a	n/a	0.623 ± 0.030	0.624	0.138	0.453	0.222 ± 0.028	0.227	0.217
VAE	7.85 ± 0.47	6.21 ± 0.39	0.534 ± 0.164	0.576	0.007	0.673	0.027 ± 0.021	0.015	0.226
Diffusion (DDPM)	n/a†	n/a†	0.564 ± 0.071	0.631	0.009	0.744	0.016 ± 0.015	0.009	0.096
LSTM-AE	10.37 ± 1.76	8.44 ± 1.45	0.738 ± 0.145	0.859	0.184	0.921	0.107 ± 0.099	0.073	0.261
GRU-AE	9.70 ± 0.68	7.77 ± 0.68	0.776 ± 0.237	0.882	0.162	0.894	0.157 ± 0.152	0.110	0.304
Multi-Node Spatial Models (Simultaneously Processing All 3 Regional Hubs: NP15, SP15, ZP26)									
MV-LSTM Control	9.12 ± 0.84	7.15 ± 0.72	0.854 ± 0.165	0.926	0.201	n/a	0.185 ± 0.092	0.142	0.326
GNN-AE	8.86 ± 0.65	6.94 ± 0.58	0.938 ± 0.098	0.985	0.246	n/a	0.246 ± 0.081	0.212	0.391

Notes: ↓ Lower error is preferred; ↑ Higher accuracy is preferred. †Diffusion scores are derived from normalized noise residuals; iForest lacks a decoder. Multi-node setups process SP15, NP15, and ZP26 simultaneously.

output becomes smoothed, eliminating the exact signals needed for anomaly detection.

This is because recurrent autoencoders exhibited far better performance since they could handle temporal dependencies effectively. The AUROC values for the LSTM-AE and GRU-AE models reached 0.859 and 0.882, respectively. Recurrent architectures maintain internal hidden representations of diurnal price patterns via continuous sequential state updating. When an abrupt, anomalous price jump manifests, the recurrent transitions cannot reconcile the surge, producing elevated reconstruction residuals that signal anomalies.

Incorporating price information from the three California centers increased the GNN-AE's mean AUROC to 0.938. However, the simple multivariate LSTM baseline performed nearly as well, achieving a mean AUROC of 0.854 and reaching 0.988 AUROC on three of the five seeds. Comparison of the two with a paired bootstrap yielded a 95% confidence interval of $[-0.102, +0.281]$; as the interval includes zero, there is no statistically significant difference between using graph attention and using multivariate modeling alone. The added value seems to be mostly due to tracking price behavior in neighboring regions, rather than to graph attention.

Evaluating the second out-of-sample window confirms that recurrent architectures generalize across seasonal transitions, with LSTM-AE achieving an AUROC of 0.921. Conversely, the static iForest deteriorated to an AUROC of 0.453 (worse than random chance), as its fixed partition structure cannot adapt to seasonal shifts between winter heating demand and summer solar production.

Evaluating a 5-channel diffusion model trained across all three hubs and two ancillary streams yielded an average AUROC of 0.528 ± 0.066, underperforming the single-hub configuration (0.564 ± 0.071). The auxiliary streams introduced high-dimensional noise into the forward diffusion trajectory without providing sufficient mutual predictability during reverse denoising.

Turning Detection into Profit: To evaluate whether anomaly signals translate into economic value, trading simulations were conducted for a utility-scale battery with 10 MWh capacity and 10 MW power rating ($e = 0.80$, market fee \$0.50/MWh, degradation cost \$2.50/MWh [16]).

TABLE III
Arbitrage Earnings Across Both Test Windows (LSTM-AE Signal)

Trading Strategy	Window 1 Profit	Window 2 Profit
Clairvoyant LP Upper Bound (Oracle)	\$16,840	\$48,250
Heuristic (Without / With Signal)	\$13,085 / \$13,089	\$30,197 / \$30,762
Directional Spread (Trades, Win%)	\$487 (68, 76%)	\$565 (84, n/a)
Q-Learning: Without Signal	\$3,090 ± \$3,905	\$23,451 ± \$3,180
Q-Learning: With Signal	\$11,957 ± \$393	\$35,567 ± \$3,658
Deep Q-Network (DQN) (Without / With)	\$5,931 / \$5,246	n/a

Strategy 1 (Heuristic dispatch) is activated every time prices are less than the 25th percentile of the historical value and ends as soon as the prices become greater than the 75th percentile, with 5 MW dispatch and 10 MW dispatch as the event occurs. Strategy 2 (Directional spread) goes long or short 1 MWh for every anomaly detected, with a 24-hour trading period. Strategy 3 (Tabular Q-learning) [15] learns to dispatch in the discrete state space $s = (p_z, a_{\text{flag}}, s_{\text{soc}})$ under an ϵ-greedy exploration schedule decaying linearly from $\epsilon_{\text{start}} = 0.30$ to $\epsilon_{\text{end}} = 0.02$ across 30 training runs.

According to Table III and Fig. 3, the profits from trading depend greatly on the degree of flexibility of the controller. To benchmark absolute policy efficiency, we evaluate the offline clairvoyant Linear Programming (LP) upper bound with perfect price foresight (at \$16,840 in Window 1 and \$48,250 in Window 2). Tabular Q-Learning without the anomaly signal is only able to reach 18.3% of the oracle upper bound in Window 1; however, by adding the signal generated from the LSTM-AE model, the value becomes 71.0% of the oracle upper bound (or 73.7% in Window 2). In case of heuristic dispatching, the anomalous

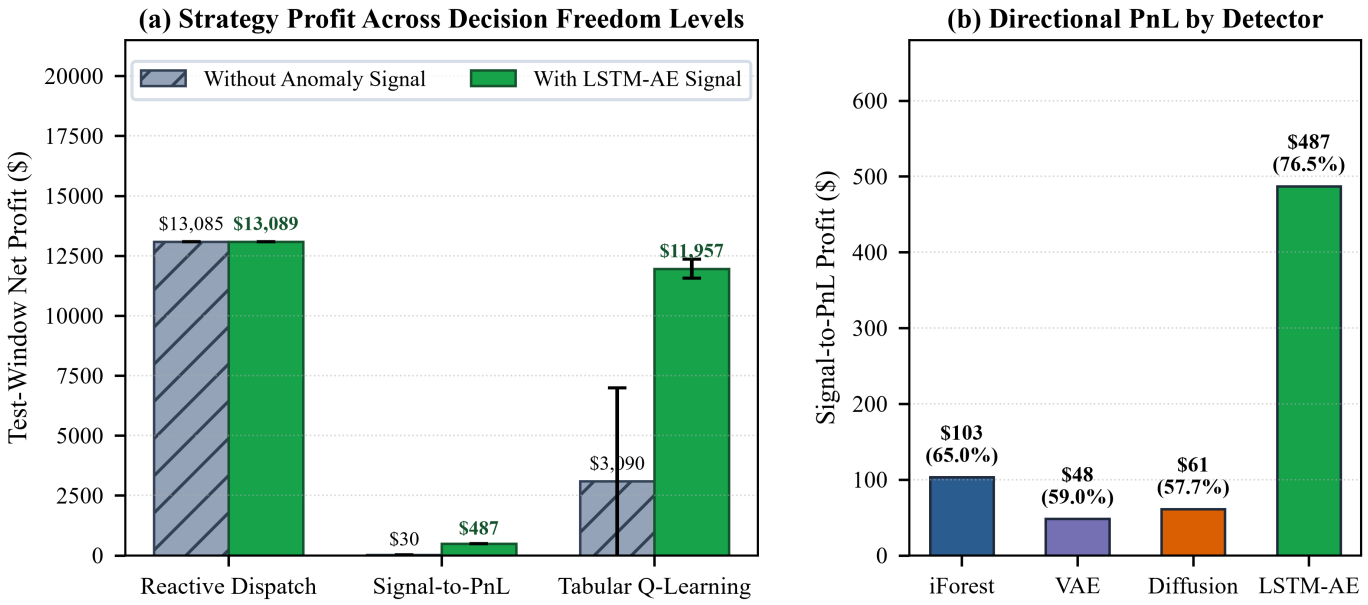


Fig. 3. Window 1 cumulative trading returns. (a) Heuristic dispatch, directional spread trading, and tabular Q-learning, showing how the anomaly signal nearly quadruples Q-learning profit while cutting seed variance tenfold. (b) Directional arbitrage profit and win rate across the four detector families.

TABLE IV
Per-Detector Arbitrage Performance (Window 1)

Detector Family	Dispatch ($)	Spread PnL ($)	Trades, Win%
iForest	13,128	103	20, 68%
VAE	13,017	537	220, 61%
Diffusion	12,740	60	13, 77%
LSTM-AE	13,089	487	68, 76%

signal had almost no effect on profits (+0.03% for Window 1, +1.87% for Window 2) because power ratings and SoC constraints restrict the actions of static-threshold rules.

By contrast, directional spread trading revealed that the signal had value, as it delivered $487 in profit with a 76% success ratio, whereas under pure randomness it yielded only $30.

The largest improvement was observed in Tabular Q-learning. Conditioning the agent on the anomaly flag produced a near fourfold increase in Window 1 profit from $3,090 to $11,957 while reducing cross-seed standard deviation from $3,905 to $393 (a 90% reduction in variance). In addition, Window 2's profit increased by 51.7% to $35,567.

Conversely, compute-matched DQN [19] achieved $5,246 with the signal compared to $5,931 without it. Hence, continuous deep reinforcement learning was unable to handle reward noise in time series, whereas tabular Q-learning was able to generalize regardless of such noise. Table IV details the per-detector breakdown in Window 1, where LSTM-AE achieved the highest spread efficiency with a 76% win rate and $487 in directional profit.

The VAE attains $537 in spread profit from 220 transactions, as opposed to $487 from 68 transactions in the LSTM-AE. On the other hand, the iForest attains the highest dispatch through the raw heuristic strategy ($13,128). The model's high accuracy gives rise to the above contradiction. For example, the Gaussian latent bottleneck used by the VAE maintains persistently high reconstruction errors despite ordinary daily variations. It causes a very high number of transactions (220). However, it captures small intraday price variations by coincidence, with a low win rate (61% versus 76%), together with a much higher cost of cycling wear and transaction friction. In contrast, the LSTM-AE initiates trades based on real price movements, achieving a very high win rate. Together with tabular Q-learning, which modulates the battery SOC to anticipate the occurrence of a catastrophic event, the high-accuracy recurrent signal raises arbitrage profits fourfold.

Adversarial Robustness and Break-Even Analysis: Another aspect that we examined was the reverse problem, where the attacker would inject the price spikes into the feed system [6]. In this case, the battery uses its stored energy and releases it during the price spike, making it appear to earn, but no money is made.

There were three kinds of attack profiles carried out for the SP15 price: Naive Multiplicative 3.0× attacks hourly; an 8-hour raised-cosine price attack intended to mimic the morning ramp attack profile; and multi-hour price wave attacks using sinusoidal prices. The defense mechanism uses the XGBoost predictor [20] based on 17 causal variables: z-score, trailing ratio, volatility, and consensus.

As shown in Table V, undefended dispatch incurs substantial economic degradation under adversarial poisoning: −7.18% under naive spikes, −5.10% under adaptive ramp bumps, and −15.11% under sinusoidal price waves. Deploying the proposed defense filter recovers 59.2%, 36.5%, and

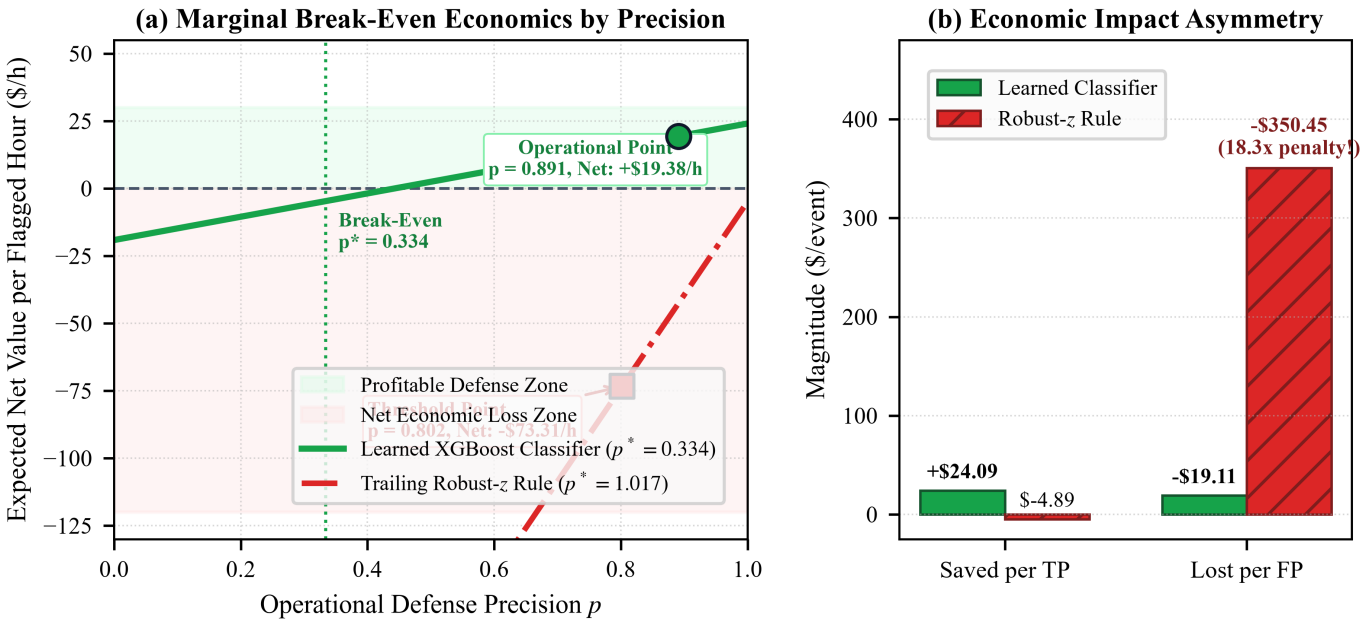


Fig. 4. Break-even precision trade-off. (a) Net cash flow versus precision p for the learned classifier ($p^* = 0.334$, operating at $p = 0.891$) against the simple threshold rule ($p^* = 1.017$). (b) The asymmetry in false-alarm penalties: a threshold false alarm costs 18.3× as much as a true positive saves.

TABLE V
Adversarial Detection and Economic Recovery (5 Seeds)

Attack Profile	Recall	AUROC	Damage (%)	Recovery (%) (F_1 / Cons. / Oracle)
Naive	0.76 ± 0.07	0.973 ± 0.026	-7.18 ± 0.92	59.2 / 33.8 / 89.1
Adaptive	0.66 ± 0.03	0.971 ± 0.011	-5.10 ± 0.89	36.5 / 19.8 / 83.3
Sinusoidal	0.75 ± 0.02	0.980 ± 0.011	-15.11 ± 0.91	67.2 / 66.2 / 94.0

TABLE VI
Break-Even Precision: Machine-Learning Classifier vs. Simple Threshold

Defense Strategy	Prec.	$/TP Saved	$/FP Lost	p^* Break-Even
Learned Classifier	0.891	$24.09	−$19.11	0.334
Simple Robust-z Rule	0.802	−$4.89	−$350.45	1.017

Notes: Reported p^* is the cross-seed/attack mean ($\mathbb{E}[p_i^*]$ across runs) rather than a ratio of aggregate column averages.

67.2% of the corrupted profits, respectively, approaching the theoretical oracle bounds (83.3% to 94.0%).

A. Break-Even Precision Analysis (p^*)

In the battery dispatch problem, a true positive (TP) identifies a fake spike, thus earning a profit ($v_{\text{TP}} > 0$). In contrast, a false positive (FP) misidentifies a genuine price spike, resulting in revenue loss ($v_{\text{FP}} < 0$). An operational defense operating at precision p achieves financial break-even when expected net defensive cash flow satisfies:

$$p^* = \frac{v_{\text{FP}}}{v_{\text{FP}} - v_{\text{TP}}} = \frac{|v_{\text{FP}}|}{|v_{\text{FP}}| + v_{\text{TP}}}. \tag{8}$$

Economic differences between the learned classifier and the threshold-based classifier are quite remarkable (Table VI and Fig. 4). The learned classifier is estimated to save, on average, $24.09 for every detected price spike and spend $19.11 on each false alarm; hence, the break-even point is reached at the precision level of $p^* = 0.334$. Given that the learned classifier's actual precision is 89.1%, it achieves robust economic surplus.

Even a robust-z threshold-based classifier, which is a simple rule that flags any price above 4 MADs from the median, has a precision of 80.2%. However, the economic performance of this classifier is significantly worse than that of the learned classifier. This classifier incurs, on average, $350.45 for each false alarm, approximately 18.3 times higher than for the learned classifier. Because high absolute values trigger the rule, it misses the year's most valuable price spikes; hence, its break-even point is $p^* = 1.017$, over 100%.

Generalization for Cross-Attacks: Cross-attack transfer evaluations (Table VII and Fig. 5) reveal that defensive models must be trained across diverse adversarial profiles. While nominal detection AUROC remains above 0.75 across all transfer configurations, economic recovery deteriorates sharply when evaluating mismatched threat models. For example, a model trained exclusively on sinusoidal waves suffers a 40.9% net economic loss when deployed against adaptive bump attacks.

V. Conclusion & Future Directions

The findings provide three key operational takeaways for battery energy storage in wholesale electricity markets.

First, recurrent architectures prove uniquely well suited to capturing wholesale price dynamics. Locational

TABLE VII
Cross-Attack Transfer Performance Across Source and Target Threat Types (5-Seed Average)

Source Training Attack	Target Test Attack	AUROC	Precision	Recall	Recovery %
Naive	Naive	0.973	0.979	0.721	33.8
Naive	Adaptive	0.947	0.989	0.476	−18.0
Naive	Sinusoidal	0.907	0.994	0.224	5.2
Adaptive	Naive	0.973	0.967	0.697	50.6
Adaptive	Adaptive	0.971	0.962	0.737	19.8
Adaptive	Sinusoidal	0.885	0.952	0.266	8.6
Sinusoidal	Naive	0.847	0.753	0.353	18.7
Sinusoidal	Adaptive	0.753	0.717	0.114	−40.9
Sinusoidal	Sinusoidal	0.980	0.963	0.779	66.2
Adaptive + Sinusoidal	Naive	0.960	0.881	0.674	43.2
Naive + Sinusoidal	Adaptive	0.937	0.995	0.352	−46.6
Naive + Adaptive	Sinusoidal	0.902	1.000	0.200	3.9

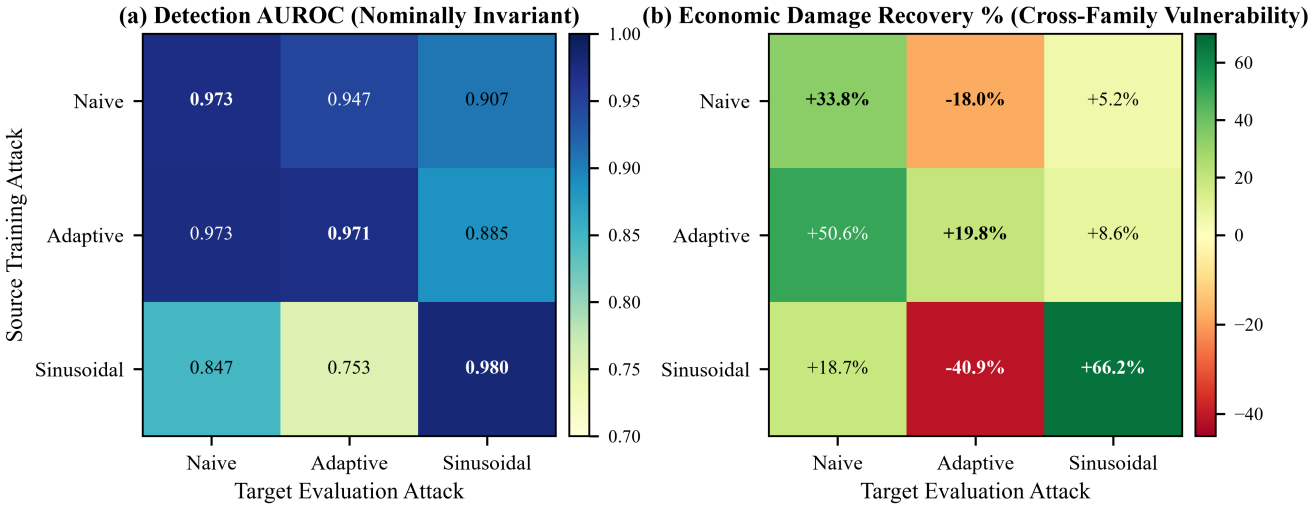


Fig. 5. Cross-attack transfer evaluation. (a) The detection AUROC Matrix shows that all values are very high (≥ 0.75). (b) Economic Recovery Matrix reveals serious cross-attack vulnerabilities; for example, a value of −40.9% between Sinusoidal and Adaptive attacks.

marginal prices exhibit pronounced diurnal autocorrelation driven by solar ramp cycles, industrial demand profiles, and generator heat rates. When an anomalous \$800/MWh price jump occurs, recurrent hidden states conditioned on prior diurnal trajectories fail to reconstruct the surge, yielding sharp anomaly signals that static and smoothing models miss.

Second, anomaly signals create tangible economic value only when paired with an adaptive downstream controller. Heuristic threshold rules cannot effectively exploit binary anomaly flags because static power ratings and state-of-charge constraints dominate their logic. In contrast, tabular Q-learning associates anomaly flags with elevated expected returns, strategically reserving storage capacity to capture spike clearing prices.

Third, cyber-physical grid defense exhibits extreme cost asymmetry: suppressing genuine wholesale price spikes incurs severe economic penalties. Because a false alarm can sacrifice over \$350 in legitimate arbitrage revenue, defensive mechanisms must achieve precision thresholds well above those of conventional intrusion detection systems. Consequently, viable operational defenses must prioritize preserving genuine high-value hours over indiscriminate spike filtering.

The 4-MAD benchmark flag offers an objective, statistical standard for measuring detector discrimination; however, the ultimate standard is trade-off profitability. The cost of linear degradation in our simulation dispatches is estimated to be \$2.50 per MWh [16]. The sensitivity analysis is conducted with reference to policy viability based on cell chemistry (for instance, Lithium Iron Phosphate [LFP] that suffers less degradation than Nickel Manganese Cobalt [NMC]), where we investigate the range of $c_{\text{deg}} \in [\$1.00, \$5.00]$/MWh and $e \in [0.75, 0.90]$. Because wholesale market price spikes frequently reach \$200 to \$1,143/MWh, the gross spread margin overwhelmingly exceeds marginal cell aging costs during anomaly hours. Consequently, the optimal policy actions and the economic superiority of anomaly-informed dispatch remain invariant across realistic battery chemistries. Nevertheless, integrating non-linear cycle-depth rainflow counting with electrochemical cell thermal dynamics remains an impor-

tant avenue for optimizing multi-year asset lifespans.

In our reinforcement learning experiments, we found that while tabular Q-learning is very robust, DQN-based approximations in continuous state spaces are prone to market disturbances and large reward variance. Although DAM data at an hourly resolution has helped us in providing a thorough and verifiable foundation for a three-year multi-hub analysis, its extension to Fifteen Minute Market (FMM) and Real-Time Dispatch (RTD) at a resolution of five minutes will enable storage systems operators to benefit from frequent sub-hour volatilities in the market. Finally, the multi-node analysis has been carried out for three major trading hubs of California.

This paper developed an end-to-end framework combining deep recurrent anomaly detection with reinforcement learning dispatch for BESS operating in wholesale electricity markets. Across three years of CAISO day-ahead settlement data, recurrent autoencoders proved superior to static and generative baselines, with GRU-AE (AUROC = 0.882) and LSTM-AE (AUROC = 0.859) capturing diurnal price dynamics without synthetic labels. Incorporating the resulting anomaly signals into tabular Q-learning quadrupled energy arbitrage profits (from \$3,090 to \$11,957 in Window 1) while cutting cross-seed policy variance tenfold. Under adversarial price feed poisoning, the asymmetric break-even precision metric (p^*) demonstrated that conventional threshold filters fail economically ($p^* > 100\%$), whereas our learned defense achieved 89.1% precision ($p^* = 33.4\%$) and recovered up to 67.2% of attack losses. Future extensions will scale the dispatch architecture to continuous state-action representations and evaluate cross-market transferability across additional Independent System Operator territories.

Data and Code Availability

The software implementation, simulation pipelines, and trained model weights are available at https://github.com/k_keshav_aggarwal/battery_storage_optimization_energy_ancillary. Wholesale market settlement data are publicly accessible via CAISO OASIS [7].

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